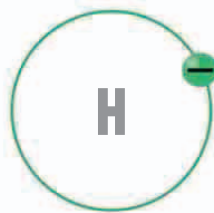


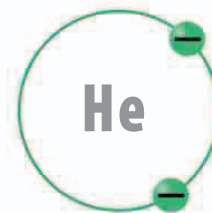
Hydrogen and helium

Hydrogen is in Group 1 and has a single outer electron. However, it is not included in the alkali metals. This is because it has a distinct set of chemical properties compared to the other group members. Similarly, helium has two outer electrons, yet it is not included in Group 2. Instead it is in Group 18 with the noble gases, with which it shares most chemical properties.



△ Hydrogen

Hydrogen has just one electron, which it loses less easily than other elements in its group.



△ Helium

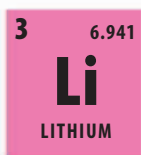
Helium has one electron shell, and with two electrons it is full. It does not form ions in chemical reactions.

Group trends

Members of Groups 1 and 2 become more reactive down the group as the atoms get bigger. This is because the atoms' negatively charged outer electrons are located farther away from the positively charged nucleus—and are held less strongly in the atom.

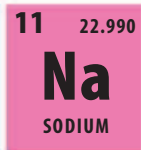
▽ Alkali earth metals

Earth metals are so called because they are found in compounds in the Earth's crust. Beryllium, for example, is found in gemstones such as emeralds. Although Group 2 metals react with water, they do so less strongly than those in Group 1.



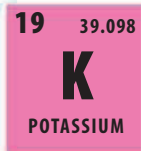
◁ Lithium

With the lowest density of any metal, lithium even floats in water.



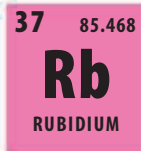
◁ Sodium

The most abundant alkali metal, sodium compounds are found in many rocks.



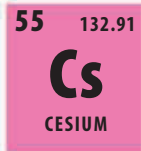
◁ Potassium

Potassium is named after potash—potassium—containing compounds in the ash of burned wood.



◁ Rubidium

This metal would melt on a hot day and is so reactive that it catches fire in air.



◁ Cesium

Cesium melts at 28°C (82°F), which is only just above room temperature.



◁ Francium

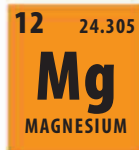
This radioactive metal is extremely rare and little is known about it.

high reactivity



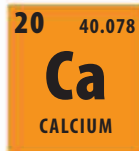
◁ Beryllium

This metal has a very low density so it is used to make high-speed aircraft and satellites.



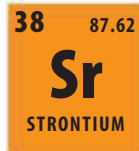
◁ Magnesium

This metal is named after the region Magnesia in Greece, which has lots of magnesium compounds.



◁ Calcium

Calcium is common in Earth's rocks. Natural calcium compounds are often known as limes.



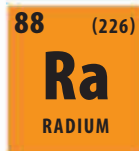
◁ Strontium

While most strontium is relatively stable, some forms are radioactive and dangerous.



◁ Barium

Barium compounds are added to fireworks to produce green explosions.



◁ Radium

This metal is highly radioactive and gives off a faint blue glow.

high reactivity